Real-Time Stochastic Assessment of Dynamic N-1 Grid Contingencies

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Introduction

Power system operators need fast, real-time tools to assess grid security under dynamic conditions, **not just static N-1 scenarios**. Conventional analyses often ignore frequency swings caused by common disturbances such as single-phase faults. We present a **real-time dashboard framework to screen dynamic contingencies** using realistic fault scenarios and transient responses. Assumptions:

- **(a)** The grid is initially in a **balanced operating state.**
- **(b) Fault is cleared within a second (or a shorter system-dependent interval).**
- **(c) A contingency is flagged** if any line exceeds **its safety threshold power flow** during the post-fault transient.

Contributions

We aim to **efficiently assess and analyze power grid dynamics under realistic fault conditions** by introducing:

- **Fast N-1 dynamic screening** to simulate transient responses to counterfactual faults on any line from a given grid state.
- **A probabilistic framework** to model the likelihood of overloads over time using phase angle dynamics to support reliability analysis under uncertainty.

Motivation

- **System Stability:** Swing equations analyze transient stability, ensuring a power system can recover from disturbances and maintain generation-consumption balance **[1].**
- **N-1 Contingency:** Ensures reliability after single failures, but is static, slow, and limited for modern, renewable-rich grids **[2]**.
- **Rare Event/Instanton Sampling:** Reveals likely causes of extreme grid failures by targeting high-impact, low-probability scenarios beyond standard Monte Carlo **[3]**.

Road map

We illustrate and summarize the constitution of this work on Fig. (1):

N-1 Plus Pipeline – to Monitor Dynamic N-1 Contingencies

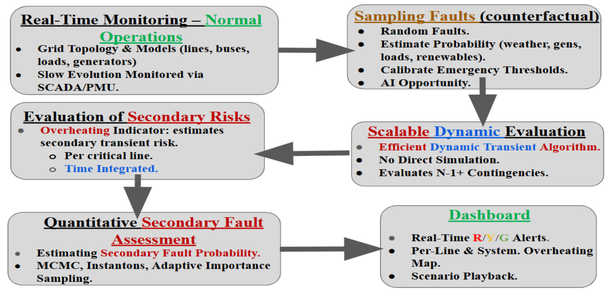

Figure 1: Conceptual dashboard pipeline for dynamic contingency screening.

Swing Equations

We consider **the swing equation in the linear approximation** :

$$m_i\ddot{\theta}_i(t) + d_i\dot{\theta}_i(t) + \sum_{\{i,j\}\in\mathcal{E}} \beta_{ij}(\theta_i(t)-\theta_j(t)) = P_i : i\in\mathcal{V} \qquad (1)$$

where $\mathcal{V}$ and $\mathcal{E}$ are the sets of nodes and (undirected) edges of the power system graph $\mathcal{G}=(\mathcal{V},\mathcal{E})$. The set of nodes $\mathcal{V}$, the vectors of internal voltage angles/phases $\theta(t)=(\theta_i(t)|i\in\mathcal{V})$, inertia $\boldsymbol{m}=(m_i|i\in\mathcal{V})$, damping $\boldsymbol{d}=(d_i|i\in\mathcal{V})$ and the power injection $\boldsymbol{p}=(P_i|i\in\mathcal{V})$ (where $\sum_{i\in\mathcal{V}}P_i=0$) stay constant through out the dynamics considered. And $\beta=\{\beta_{ij}|\{i,j\}\in\mathcal{E}\}$ represents the set effective (scaled) **stiffness** of the grid. **We will also define the safety polytope:**

$$\Pi_{\boldsymbol{\theta}} \doteq \left\{\boldsymbol{\theta}\in[-\pi,\pi]^{|\mathcal{V}|}\middle|\forall\{i,j\}\in\mathcal{E}, \left|\beta_{ij}\left(\theta_i-\theta_j\right)\right|\leqslant\bar{p}_{ij}\right\}. \qquad (2)$$

The problem is initialized at the **steady state:**

$$\sum_{\{i,j\}\in\mathcal{E}} \beta_{ij}(\theta_i(0)-\theta_j(0)) = P_i,\ \ddot{\theta}_i(0)=\dot{\theta}_i(0)=0 : i\in\mathcal{V}, \qquad (3)$$

that is Eq. (1) reduces to the **static Power Flow (PF)** equations so that $\boldsymbol{\theta}(0)\doteq(\theta_i(0)|i\in\mathcal{V})\in\Pi_{\boldsymbol{\theta}}$.

Simulating Faults [4]

The linearized swing equations (1) governing phase dynamics from fault onset $(t=0)$ to post-fault stabilization $(t=T)$ – including the fault duration τ – can be expressed compactly as:

$$\dot{\boldsymbol{x}} = \boldsymbol{A}(t)\,\boldsymbol{x}+\boldsymbol{P}, \qquad (4)$$
$$\boldsymbol{P}\doteq\begin{pmatrix}\boldsymbol{p}\\ \boldsymbol{0}\end{pmatrix},\ \boldsymbol{x}\doteq\begin{pmatrix}\dot{\boldsymbol{\theta}}(t)\\ \boldsymbol{\theta}(t)\end{pmatrix},\ \boldsymbol{x}_0\doteq\begin{pmatrix}\boldsymbol{0}\\ \boldsymbol{L}_0^{-1}\boldsymbol{p}\end{pmatrix},$$
$$\boldsymbol{A}(t)=\begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{D} & \boldsymbol{M}^{-1}\boldsymbol{L}(t)\\ \mathbb{O}_{n\times n} & 1_{n\times n}\end{bmatrix}=\boldsymbol{A}_0+\mathbb{I}\Big(t\in[0,\tau]\Big)\,\delta\boldsymbol{A}.$$

Where M,D : diagonal inertia and damping matrices; $L(t)$: time-varying Laplacian due to fault-induced changes in **line $\{i,j\}$'s susceptance**: **Three-Phase Faults**: line is de-energized, i.e. $\beta_{ij}\to 0$, **Single-Phase-to-Ground Faults**: susceptance is partially reduced by one third, i.e. $\beta_{ij}\to\frac{2}{3}\beta_{ij}$; $A_0,\delta A$: pre-fault dynamics and fault-time correction.

Measuring the degree of line overload

To assess line overload severity during transients, we define an **overload indicator** using the solution of the linearized swing equations (1). Let $\theta(t)=(\theta_i(t)|i\in\mathcal{V})$ denote the vector of phase angles over $[0,T]$, and $\beta_{ij}(t)$ the time-varying susceptance of line $\{i,j\}$. The line-specific overload indicator for $\{i,j\}$ is:

$$S_{ij}(\theta_{0\to T})\doteq\int_0^T\mathbb{I}\Big(\Big|\beta_{ij}(t)\big(\theta_i(t)-\theta_j(t)\big)\Big|>\bar{p}_{ij}\Big)\mathrm{d}t, \qquad (5)$$

Here, $\bar{p}_{ij}$ is the thermal threshold for line $\{i,j\}$. To evaluate the overall system stress over the set of monitored lines $\mathcal{E}_{\mathrm{m}}\subseteq\mathcal{E}$, we define a **global overload indicator:**

$$S(\theta_{0\to T})\doteq\sum_{\{i,j\}\in\mathcal{E}_{\mathrm{m}}} S_{ij}(\theta_{0\to T}). \qquad (6)$$

Probabilistic Risk Assessment [5]

To evaluate the risk of exceeding the safety region $\Pi_{\boldsymbol{\theta}}$ we use the overload indicator $S(\theta_{0\to T})$ combined with **Cross Entropy (CE)**[1] sampling. Let $x^{(0)}_{0\to T}=x(0\to T)$ be the nominal trajectory solving Eq. (4) and $S_{ij}\left(\theta^{(0)}_{0\to T}\right)$ be its line-specific overload. For a threshold $\gamma>0$, we define Q_{ij} : **the overload probability of line $\{i,j\}\in\mathcal{E}$ and $\hat{Q}_{ij}$: its respective (CE) estimate**, i.e.

$$Q_{ij}=\mathbb{P}\left[S_{ij}(\theta_{0\to T})\geqslant\gamma\right]=\mathbb{E}\left[\mathbb{I}\left(S_{ij}(\theta_{0\to T})\geqslant\gamma\right)\right],$$
$$\hat{Q}_{ij}=\frac{1}{N}\sum_{k=1}^{N}\mathbb{I}\Big(S_{ij}\left(\theta^{(k)}_{0\to T}\right)\geqslant\gamma\Big)\frac{p_Z(\alpha_k,\tau_k)}{q^{\star}_Z(\alpha_k,\tau_k)}{}^{2},$$

s.t. $(\alpha_k,\tau_k)\sim q^*_Z$; $\boldsymbol{x}^{(1)}_{0\to T},\ldots,\boldsymbol{x}^{(N)}_{0\to T}$ sample trajectories.

Case Study 1: Israeli Power Grid

We apply our **Probabilistic Risk Assessment** to an open source model of the **Israel Electric Corporation (IEC) Power Grid** consisting of 36 buses, 32 transmission lines, and 11 generators. The details and the required parameters of this model are outlined and can be found in **[6]**.

Statistics of Three Phase Fault Simulation

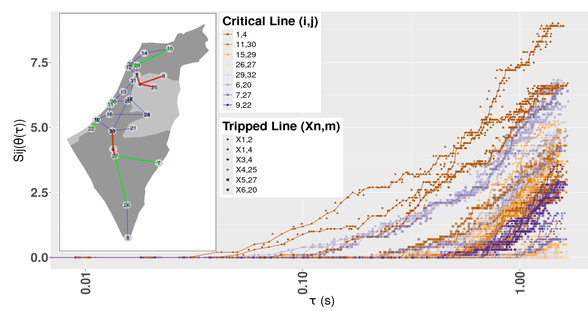

Figure 2: Overload indicator vs. fault duration (3-Phase Faults) · Each curve shows line-specific overload vs. fault time τ. **Tripped lines** $(X_{n,m})$: trigger peak overload **Critical lines (associated with transformers)** (i,j): **accumulate** highest total overload. Left: Israel grid with **relevant lines** highlighted.

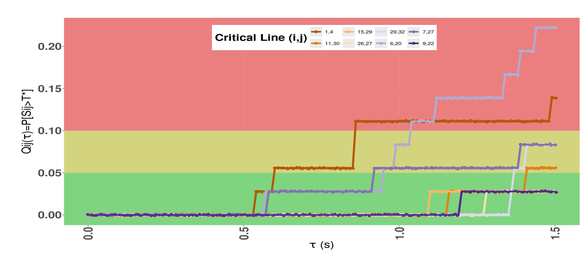

Figure 3: Probabilistic security assessment (3-Phase Faults). For critical lines and fault durations τ we determine $\mathbb{P}\left[S_{ij}\geqslant T^{(*)}\right]$. Risk zones by $Q_{ij}(\tau)<5\%$: **Safety Zone**, $Q_{ij}(\tau)\in[5\%,10\%]$: **Warning Zone** and $Q_{ij}(\tau)>10\%$: **Emergency Zone**.

[1]The (CE) method is an **adaptive importance-sampling** technique that iteratively adjusts a sampling distribution so that **rare or high-value events become much more likely under that distribution**.
[2] $p_Z(\alpha_k,\tau_k)$ and $q^{\star}_Z(\alpha_k,\tau_k)$ are the **real and "elite" distributions** of **tripped lines and fault's duration**.

Statistics of Single Phase Fault Simulation

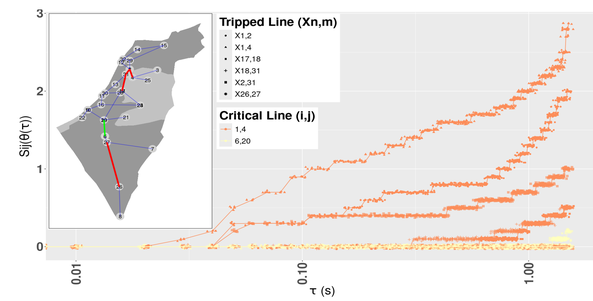

Figure 4: Overload Indicator vs. Duration of Fault for the case of single phase faults, otherwise notations and descriptions are the same as in Fig. (2)

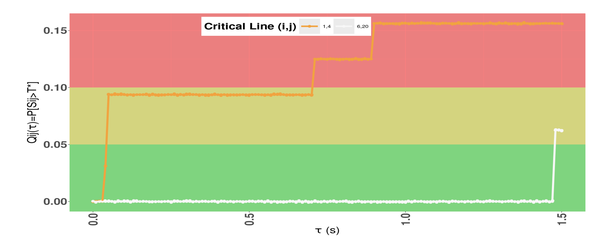

Figure 5: Probabilistic security assessment for the case of single phase faults and critical lines shown in Fig. (4), otherwise notations and descriptions are the same as in Fig. (3).

Benchmark Analysis

γ [s]	Sample Size	$\hat{Q}\approx\mathbb{P}[S>\gamma]$	Mean Time [s]
Cross Entropy Method			
0.5	2500	**0.07538**	6.749×10^{1}
5.0	5000	**0.00319**	1.296×10^{2}
10.0	10000	**0.00042**	2.554×10^{2}
Monte Carlo Method			
0.5	25000	**0.07401**	6.283×10^{2}
5.0	25000	**0.00319**	6.283×10^{2}
10.0	50000	**0.00041**	1.259×10^{3}

Figure 6: Processing Time & Convergence: CEM vs. MC for overload probability at threshold γ

References

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